

SIMPLE MATH

Scientific documents with L^AT_EX course

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Suppose you want to include some math in your documents. Just follow these basics steps:

1. Create a new T_EX document with the basic requirements (the documentclass and begin and end document commands).
2. Save, generate your pdf, and open it.
3. First let's try to add some math mid sentence.

- (a) In your document, type

Try this!

The equation of a line is $y=mx+b$ where m is the slope and b is the y intercept.

- (b) Recompile to pdf

- You should get: The equation of a line is $y=mx+b$ where m is the slope and b is the y intercept.
- This looks okay, but could be *much* better. To tell L^AT_EX that certain parts of this sentence are math, we must surround them with \$.

- (c) Insert \$ around all math:

Try this!

The equation of a line is $y=mx+b$ where m is the slope and b is the y intercept.

- (d) Recompile to pdf

- You should get: The equation of a line is $y = mx + b$ where m is the slope and b is the y intercept.
- Switches to the standard italicized math font and will properly handle spacing in equations automatically.

4. Of course, you may have more complicated equations that deserve their own line. For this, we use the align environment.

- (a) The align environment will require the amsmath package

- Somewhere before `\begin{document}`, you will need to type `\usepackage{amsmath}`. (This package is so commonly used that I just include it in all my T_EX documents.)

- (b) In your document, type

Try this!

The equation of a line is
`\begin{align}`
 $y=mx+b$
`\end{align}`
where m is the slope and b is the y intercept.

(c) Recompile. You will get: The equation of a line is

$$y = mx + b \tag{1}$$

where m is the slope and b is the y intercept.

(d) We can also line up the equal signs in multiple equations (or any other point in the equations), by inserting a \$;

- In your L^AT_EX document, type:

```
\begin{align}
x&=1+1+1+1\\
&=4.
\end{align}
```

Try this!

- This will produce:

$$x = 1 + 1 + 1 + 1 \tag{2}$$

$$= 4. \tag{3}$$

Notice that the & does not appear in either equation, but the place where the & would go is nicely aligned.

5. You may also want to put equations on a new line and have the alignment capabilities provided by the align environment, but do not want your equations to be numbered. L^AT_EX can handle this also. There are two main ways to do it:

- At the end of your equation inside an align environment, use the command `\nonumber`. This will remove the number for the preceding equation

- In your L^AT_EX document, type

Here is a numbered and unnumbered equation together:

```
\begin{align}
y&=mx+b\nonumber\\
z&=nw+c.
\end{align}
```

Try this!

- This produces: Here is a numbered and unnumbered equation together:

$$\begin{aligned} y &= mx + b \\ z &= nw + c. \end{aligned} \tag{4}$$

Notice that the second equation has a number, but the first does not.

- The alternative is to use the align* environment. This works exactly the same as the align environment but does not number the equations.

- In your L^AT_EX document, type

Here are two unnumbered equations together:

```
\begin{align*}
y&=mx+b\\
z&=nw+c.
\end{align*}
```

Try this!

- This produces: Here are two unnumbered equations together:

$$\begin{aligned} y &= mx + b \\ z &= nw + c. \end{aligned}$$

Notice that both equations are unnumbered.

6. You may also want to use some symbols in your equations. This is where L^AT_EX has an advantage over programs like Microsoft Word. Note that all of the following symbols must be placed inside a math environment, either between \$ or in an align environment.

- Subscripts and superscripts:

- To produce subscripts or superscripts, we precede the subscripted (superscripted) character with a `_` (`^`).
- In your document, type `$a_1^2$` to get a_1^2 .
- One thing to be careful of: `_` and `^` will only subscript or superscript *one* character. To produce something like a_{12}^{34} then, we would need to type `$a_{12}^{34}`. The brackets tell L^AT_EX to consider everything inside them as one character.

Try this!

- Greek letters:

- Command for Greek letters is their name beginning with a lowercase for the lowercase version, and a capital for the capital version.
- In your document, type
My favorite Greek letter is `$\Gamma$`, or `$\gamma$`.
to produce “My favorite Greek letter is Γ , or γ .”
- Note: some Greek letters look exactly the same as the Latin or Roman letter (the ones you use every day). For example, the capital version of β is just B and there is no corresponding L^AT_EX command `\Beta`. Just type B or the equivalent for these letters.

Try this!

- Operator symbols:

- L^AT_EX can also produce commonly used math symbols like $\int$, $\sum$, and $\prod$.
- These large operators also modify the behavior of the subscript and superscript commands to produce lower and upper bounds.
- In your document, try
Here are some random math symbols:
`\begin{align}`
`\int_0^1 \sum_a^b \prod_\alpha^\beta`
`\end{align}`
- This should produce: Here are some random math symbols:

Try this!

$$\int_0^1 \sum_a^b \prod_\alpha^\beta. \quad (5)$$

- Common functions:

- You may have noticed that L^AT_EX makes all characters inside a math environment italicized. But function names like $\sin$ and $\lim$ should not be set in italics. Fortunately, L^AT_EX has built-in commands for these as well.
- In your document, type
Here is a basic limit:
`\begin{align}`
`\lim_{x \rightarrow 0} \sin(x) = 0`
`\end{align}`
- This produces: Here is a basic limit:

Try this!

$$\lim_{x \rightarrow 0} \sin(x) = 0. \quad (6)$$

- Other:

- L^AT_EX’s selection of symbols is enormous and we cannot possibly cover all of them here, but there are a few that are common (such as the $\rightarrow$ (`\rightarrow`) in the previous example).

- Fractions:

* In your document, type

Here is a fraction:

```
\begin{align*}
\frac{12}{34}.
\end{align*}
```

* This produces: Here is a fraction:

$$\frac{12}{34}.$$

* Notice that everything in the first set of brackets goes in the numerator and everything in the second set goes in the denominator. These are the “arguments” to the `\frac{}{}` command. More on this in the “defining macros” section.

- Calculus: `\nabla` and `\partial` will produce ∇ and ∂ , respectively. Try it in your own document.

- Physics: `\hbar` will produce $\hbar$; `\vec{x}` will produce $\vec{x}$; `\mathbf{x}` will produce $\mathbf{x}$. Try these now.

Try this!

Try this!

Try this!

7. Referencing equations:

- Eventually, as your documents get longer, you will want to refer to earlier equations. One could manually compile the document to pdf, see what the equation number is, edit the T_EX file, recompile and check, but this is time consuming, especially if the equation numbers get changed later on.
- Better solution: automatic numbering. Those numbers L^AT_EX adds after your equations aren’t just there for decoration. They can be reference automatically.
- Try typing the following into your document:

The equation

```
\begin{align}
```

```
y=x
```

```
\label{mylabelledequation}
```

```
\end{align}
```

is equation `\ref{mylabelledequation}`, or just `\eqref{mylabelledequation}`, in my document.

- The example should produce: The equation

$$y = x \tag{7}$$

is equation 7, or just (7), in my document.

- *Important:* After changing labels, you will need to compile your document *twice* to get the correct output. The first time L^AT_EX figures out all the labels. The seconds time through it figures out the references. (Roughly speaking.)
- Note:
 - If you want to reference an equation, make sure it is not marked with `\nonumber` or in an `align*` environment. (You’ll want the equation to be numbered so you can reference the equation number.)
 - `\ref{}` can also be used with figures, table, etc. (see later)

Try this!

- `\eqref{}` adds parenthesis around equation numbers. This is the preferred way of referencing equations.
- The package `hyperref` can make references in the pdf document into links to the equation. The package `hyperref` can make references in the pdf document into links to the equation.

```

\usepackage{xcolor}
\definecolor{bluecite}{HTML}{0875b7}
\usepackage[unicode=true,
bookmarksopen={true},
pdffitwindow=true,
colorlinks=true,
linkcolor=bluecite,
citecolor=bluecite,
urlcolor=bluecite,
hyperfootnotes=false,
pdfstartview={FitH},
pdfpagemode= UseNone]{hyperref}

```